

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                       | (Level/CO) | Marks     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                           |            | <b>12</b> |
| A) Define pattern. Explain any four types of patterns.                                                                                                | CO1        | 6         |
| B) What is the function of core? What is core venting?                                                                                                | CO1        | 6         |
| C) Explain the investment casting process with a sketch. Why is the investment casting process capable of producing fine surface details on castings? | CO1        | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                            |            | <b>12</b> |
| A) Explain any three examples of rolling stand arrangements.                                                                                          | CO2        | 6         |
| B) Briefly explain about open die forging and closed die forging.                                                                                     | CO2        | 6         |
| C) What are the different types of defects may occur in extrusion and drawing operations?                                                             | CO2        | 6         |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                           |            | <b>12</b> |
| A) How cutting takes place when an oxyfuel gas torch is used? Also give its applications.                                                             | CO3        | 6         |
| B) Explain the Gas Metal-arc Welding (MIG welding) with a neat sketch.                                                                                | CO3        | 6         |
| C) Describe the functions and characteristics of electrodes. What functions do coatings have?                                                         | CO4        | 6         |
| <b>Q.4 Solve Any Two of the following.</b>                                                                                                            |            | <b>12</b> |
| A) What are the different machining operations that can be performed on a lathe machine? Describe them with neat sketches.                            | CO3        | 6         |
| B) Distinguish between the drilling and boring operations. Explain how can we perform both operations on a lathe?                                     | CO4        | 6         |
| C) What are the different types of chips generated in turning operations?<br>Which type of chips are desirable?                                       | CO4        | 6         |

|             |                                                                                                             |            |           |
|-------------|-------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                      |            | <b>12</b> |
| <b>A)</b>   | What is the difference between end milling and face milling operations?<br>Explain them with neat sketches. | <b>CO5</b> | <b>6</b>  |
| <b>B)</b>   | What do you mean by broaching operation? How it is done?                                                    | <b>CO5</b> | <b>6</b>  |
| <b>C)</b>   | What are the different processes used for gear finishing? Explain them.                                     | <b>CO6</b> | <b>6</b>  |

**\*\*\* End \*\*\***

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